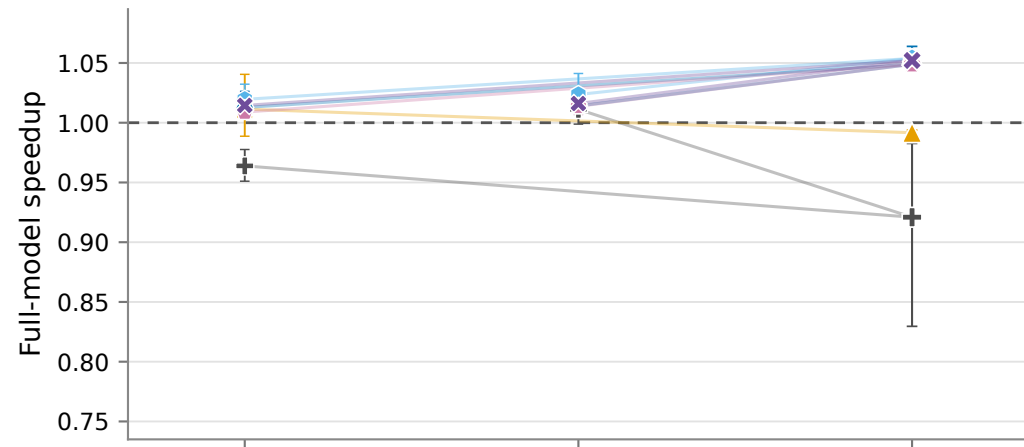


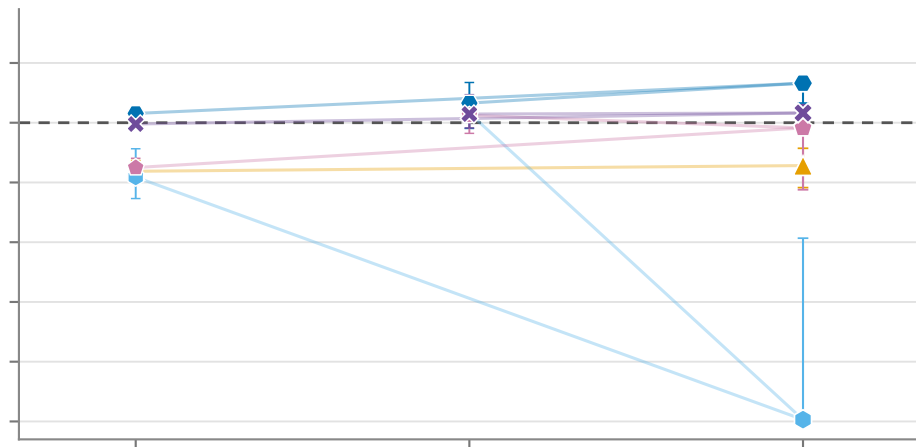
Projection and FFN eligibility contribute differently to end-to-end speed

-+- A0 ● A4-OL1, $\kappa = 0$ ● A7-OL1, $\kappa = 0$
 ▲ A1-H ● A4-OL1, $\kappa = 0.5$ × A7-OL1, $\kappa = 0.5$

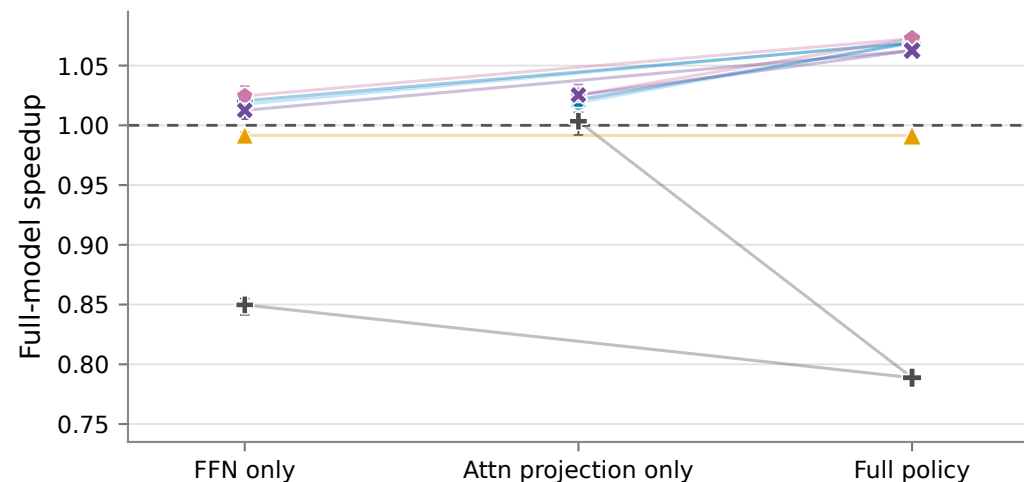
RTX PRO 4500 Blackwell - Pythia-14M



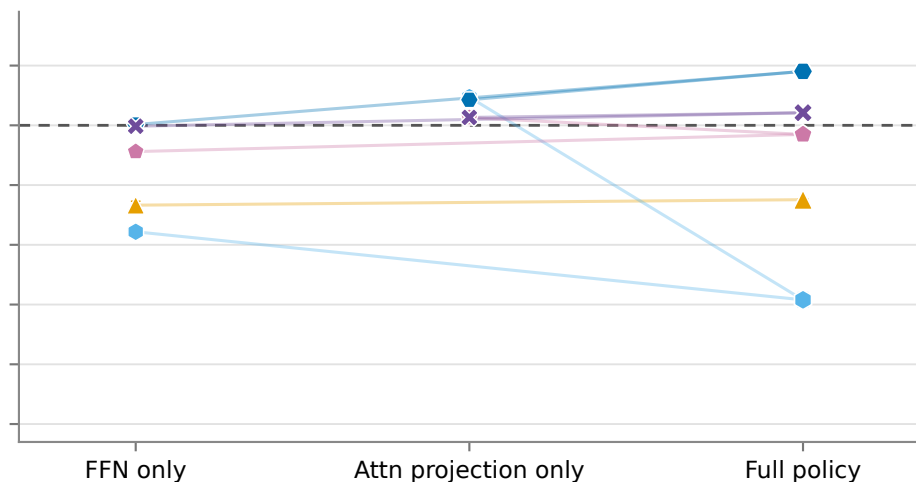
RTX PRO 4500 Blackwell - Pythia-70M



H100 NVL Hopper - Pythia-14M



H100 NVL Hopper - Pythia-70M



Component points are single-process full-model timings (bars: 95% input-cluster interval); full-policy points are process medians (bars: process range).
 Lines aid matching and do not imply additive effects. QK/PV remain dense SDPA. K010 at 410M is already z-projection-only and is omitted.